EVANS NYANNEY

Ohio University | Department of Industrial and Systems Engineering | Athens, OH
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Research Interests

Advanced Manufacturing: ◇ 3D point-cloud analysis • Geometric measurement • Reverse engineering • Additive manufacturing
Statistics & Data Science: ◇ Applied statistics • Machine learning • Optimization • Quantization • Uncertainty quantification
Healthcare Analytics: ◇ Clinical EEG • Biomedical signal processing • Intraoperative neuromonitoring

Education

2024–Present ◇ **Ohio University**, Athens, OH
Ph.D. in Industrial & Systems Engineering
GPA: 4.00/4.00 | Expected Spring 2028

2022–2024 ◇ **The University of Texas Rio Grande Valley (UTRGV)**, Edinburg, TX
M.Sc. in Mathematical Sciences (Applied Statistics & Data Science)
GPA: 3.9/4.00
Thesis: *Conditional Optimal Sets and the Quantization Coefficients for Some Uniform Distributions*
Advisor: Prof. Mrinal Kanti Roychowdhury

2012–2016 ◇ **Kwame Nkrumah University of Science and Technology**, Kumasi, Ghana
B.Sc. in Mathematics & Actuarial Science

Research Experience

Fall 2024–Present ◇ **Graduate Research Assistant**
Ohio University, Athens, OH

3D Point-Cloud and Manufacturing Research
Developing statistical methods for 3D point-cloud registration and geometric uncertainty characterization in reverse engineering and additive manufacturing quality assessment.
Collaborating with Vitruvian Building on point-cloud methods for concrete printing, geometric measurement, and quality assessment.

Clinical EEG Research
Developing and evaluating statistical and machine learning methods in collaboration with the University of Pittsburgh Medical Center (UPMC) for EEG artifact detection and interval-level detection of intraoperative cerebral ischemia.

Publications

Peer-Reviewed Journal Articles

2026 ◇ **Nyanney, E.**, Thirumala, P. D., Visweswaran, S., & Geng, Z.
A Lightweight Deep Convolutional Neural Network for Detecting Artifacts in Continuous EEG Signals. *Clinical Neurophysiology Practice*, **11**, 208–215.
doi:10.1016/j.cnp.2026.03.005

2025 ◇ **Nyanney, E.**, & Geng, Z.
Unveil the Relationship Between Process and Design Embedded in the 3D Point Cloud Using Unsupervised Learning. *Manufacturing Letters*, **44**, 1498–1506. Special Issue: NAMRC 53.
doi:10.1016/j.mfglet.2025.06.169

2025 ◇ **Nyanney, E.**, Pandey, M., & Roychowdhury, M. K.
Conditional Optimal Sets and the Quantization Coefficients for Some Uniform Distributions. *Mathematics*, **13**(15), 2350.
doi:10.3390/math13152350

2025 ◇ Hamilton, C., **Nyanney, E.**, Pandey, M., & Roychowdhury, M. K.
Conditional Constrained and Unconstrained Quantization for Uniform Distributions on Regular Polygons. *Real Analysis Exchange*, Advance Publication, 1–45.
doi:10.14321/realanalexch.1725509532

Conference and Proceedings Papers

2026 ◇ Alhawari, O., Bhutta, M. K. S., **Nyanney, E.**, & Seeharit, S.
Demand Considerations in Local and Global Markets in a Supply Chain Under Tariffs. Accepted for publication in *Lecture Notes in Production Engineering*, Proceedings of the 28th International Conference on Production Research (ICPR 28).

2023 ◇ **Nyanney, E.**, Chowdhury, J. I., Philips, D., & Lindstrom, M. R.
"Ice in the Rough: Isolating Neutron Scattering Intensities of Ice." In *Proceedings of the 7th Annual Smoky Mountain Computational Data Challenge (SMCDC)*.
https://osf.io/qgja6

Preprints

2026 ◇ **Nyanney, E.**, Thirumala, P. D., Visweswaran, S., & Geng, Z.
Detection-Guided Artifact Removal for Clinical EEG: A Selective Correction Framework.
doi:10.64898/2026.02.12.26346128

Manuscripts in Preparation

In preparation ◇ **Nyanney, E.**, & Geng, Z.
Minimax Rigid Registration for 3D Point Clouds.

In preparation ◇ **Nyanney, E.**, & Geng, Z.
A 3D Point-Cloud Structurization Algorithm.

In preparation ◇ **Nyanney, E.**, & Geng, Z.
Distributional Inference Following Minimax Registration of 3D Point Clouds.

In preparation ◇ **Nyanney, E.**, et al.
Interval-Level Detection of Intraoperative Cerebral Ischemia from EEG.

Selected Academic Projects

Spring 2025 ◇ **MyValorAI: Enhancing CoacAI with Aggression Control and Nash-Inspired Combat Decisions**
Mentor: Dr. Chang Liu
CS 4900 – Special Topics in Computer Science (GenAI)
Extended the microRTS agent CoacAI with a tunable aggression parameter and Nash-inspired combat decision rules. In a 100-game tournament, achieved an 80% win rate with approximately 12% fewer unit losses.

Fall 2023 ◇ **Context-Driven Hybrid Movie Recommendation Systems: Application to the Netflix Prize for Big Data Benchmark**
Mentor: Prof. Hansheng Lei
CYBI 6378 – Statistics and Data Analysis with Python
Explored SVD++ and cosine similarity to improve movie recommendations using Netflix data.

Spring 2023 ◇ **Neural Collaborative Filtering for Product Recommendation**
Mentor: Dr. Dongchul Kim
CSCI 6379 – Deep Learning for Neural Networks
Evaluated neural collaborative filtering on Amazon data to model implicit user–item interactions. Improved recommendation accuracy by tuning the negative sampling rate, embedding size, and network depth.

Honors and Awards

2026–2027 ◇ **IISE Future Faculty Fellow (3F)**
Institute of Industrial and Systems Engineers (IISE).

May 2026 ◇ **NSF-Supported Student Travel Award**
2026 IISE Annual Conference & Expo, Arlington, TX.

May 2024 ◇ **Honor Roll for Academic Excellence**
The University of Texas Rio Grande Valley.

February 2024 ◇ **Best Poster Presentation**
First Florence Nightingale Day, UTRGV. "Ice in the Rough: Isolating Neutron Scattering Intensities of Ice."

Summer 2023 ◇ **Best Paper**
Smoky Mountains Computational Sciences and Engineering Conference (SMCDC 2023), Challenge 2 "Diamonds & Ice: When is H_2O No Longer a Molecule," organized by Oak Ridge National Laboratory.
"Ice in the Rough: Isolating Neutron Scattering Intensities of Ice."

Teaching Experience

Summer 2024 ◇ **Graduate Teaching Assistant, Instructor of Record**
Elementary Statistical Methods (MATH 1342), UTRGV
Instructed undergraduate students in probability, statistical inference, and elementary statistical methods.

Fall 2022–Spring 2024 ◇ **Graduate Teaching Assistant, Tutor**
UTRGV Mathematics and Science Academy
Taught Calculus I, Calculus II, and Linear Algebra to high school students.

Summer 2023 ◇ **Graduate Teaching Assistant, Research Evaluator**
UTRGV High Scholars Symposium
Evaluated submitted research papers and posters.

Professional Presentations

- August 2026 — **Clinical Evaluation of a Machine Learning Model for Interval-Level Detection of Intraoperative Cerebral Ischemia During Carotid Endarterectomy**
  Poster Presentation, Clinical Abstracts Track, Machine Learning for Healthcare (MLHC 2026), Johns Hopkins Medical Campus, Baltimore, MD, August 12–15.
- May 2026 — **Detecting the Invisible: Improving Quality in 3D Manufacturing**
  3-Minute Thesis Pitch Competition, 2026 IISE Doctoral Colloquium, IISE Annual Conference & Expo, Arlington, TX.
- July 2025 — **Demand Considerations in Local and Global Markets in a Supply Chain Under Tariffs**
  28th International Conference on Production Research (ICPR 28), Universidad de La Sabana, Chía, Colombia.
- June 2025 — **Unveil the Relationship Between Process and Design Embedded in the 3D Point Cloud Using Unsupervised Learning**
  53rd NAMRI/SME North American Manufacturing Research Conference (NAMRC 53), Clemson University, Greenville, SC.
- June 2025 — **IoT-Enabled Demand Forecasting in Supply Chain Management: A Bibliometric Analysis and Future Directions**
  10th North American Conference on Industrial Engineering and Operations Management (IEOM), Orlando, FL.
- August 2023 — **Ice in the Rough: Isolating Neutron Scattering Intensities of Ice**
  Oak Ridge Smoky Mountains Computational Science & Engineering Conference, Knoxville, TN.
- April 2023 — **Quantization for the Mixtures of Overlap Probability Distribution**
  7th Annual Coastal Bend Mathematics and Statistics Conference (CBMSC), Brownsville, TX.
- April 2023 — **Life and Contribution of Erich L. Lehmann**
  Student Statistics Research Conference (SSRC), UTRGV, Edinburg, TX.

Service and Leadership

- AY 2025–Present — **Vice President**
  Alpha Pi Mu, Industrial Engineering Honor Society, Ohio University.
- 2023–2024 — **Graduate Senator**
  Student Government Association, University of Texas Rio Grande Valley.
- 2023–2024 — **Financial Committee Chair**
  Student Government Association, University of Texas Rio Grande Valley.
- July 2025 — **Reviewer**
  28th International Conference on Production Research (ICPR 2025), Chía, Colombia.

Computer Skills

- Programming Languages — Python, R, Julia, MATLAB, SAS.
- Machine Learning / GPUs — PyTorch, CUDA (working knowledge), GPU computing.
- Statistical / Simulation — RStudio, Minitab, JMP, SPSS, Simio.

Professional Memberships

- Institute of Industrial and Systems Engineers (IISE), Student Member
- American Mathematical Society (AMS)
- American Statistical Association (ASA)
- International Society for Optics and Photonics (SPIE)

References

Zhaohui Geng
Assistant Professor
Industrial and Systems Engineering
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Relationship: Primary Advisor
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